HCI Lab 2

TransitScope Malaysia

A human-centred public transport monitoring dashboard

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Problem and project objective

Public transport information remains distributed across different services.

- Commuters may switch between operator apps, sites and maps.
- Inconsistent information makes mode comparison difficult.
- TransitScope consolidates available transport records into one monitoring interface.
- The dashboard makes data mode and freshness visible before a user acts.

Users and requirements

The design prioritises daily commuters while supporting students and observers.

User role	Primary goal
Daily commuter	Locate a service and check freshness
University student	Compare modes and locate shared bikes
Transport observer	Inspect records, source and availability

- Eight functional requirements cover search, filters, map, details, alerts, charts and reset.
- Seven usability requirements address responsive use, feedback, recovery, keyboard access and source transparency.

Predefined user tasks

The study preserves three fixed tasks and time limits.

Task	Goal	Correct outcome	Limit
A	Locate a bus	Find DEMO-BUS-101 or D100 and state location and freshness	60 sec
B	Filter services	Show only shared-bicycle records on map and list	45 sec
C	Investigate alert	State severity, meaning, affected service and timestamp	60 sec

No task definition changes after testing. Demonstration records keep the tasks available without external data.

Architecture and data status

One typed model drives every dashboard view.

Future provider adapters
GTFS, GTFS-Realtime, GBFS

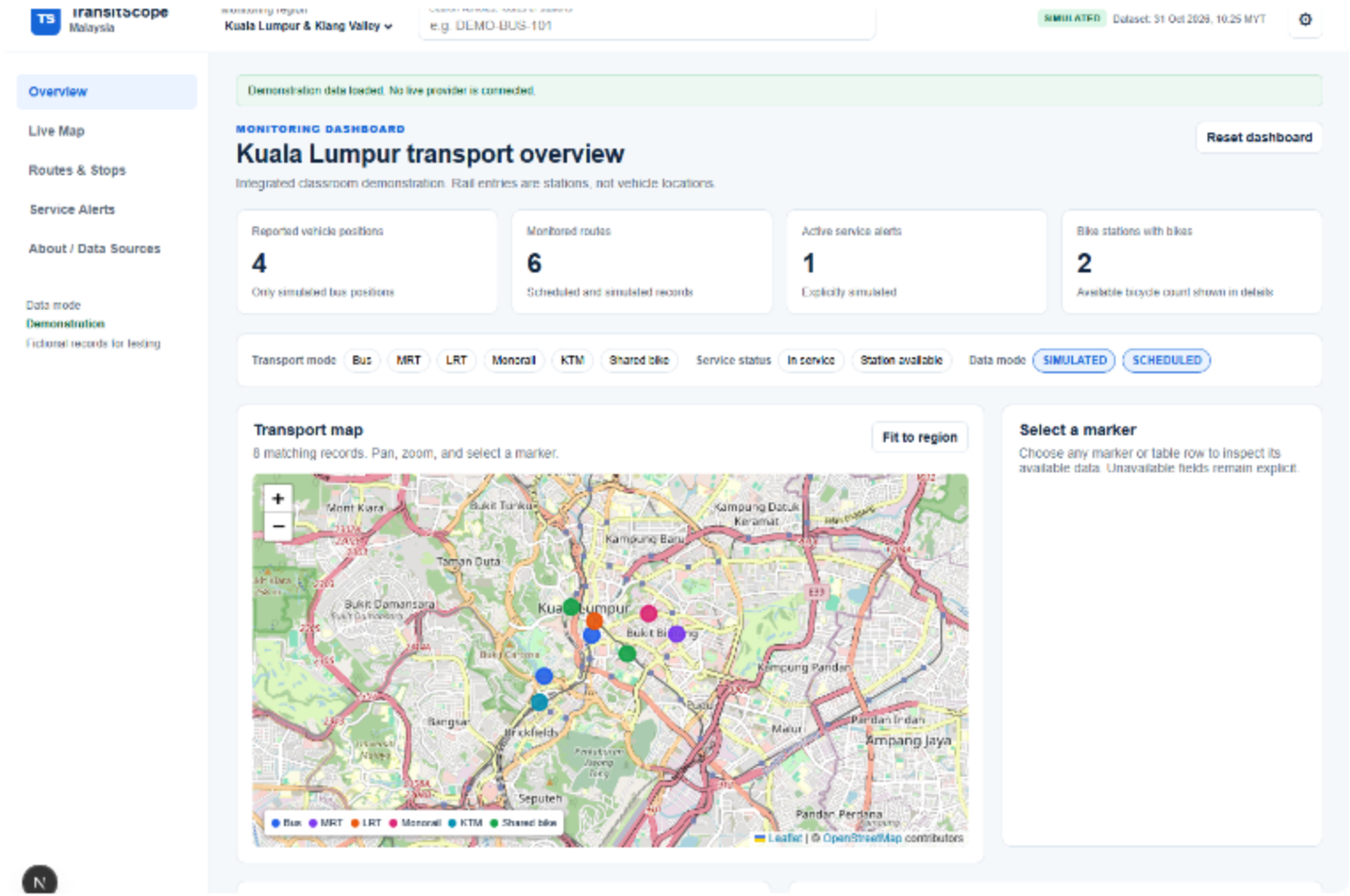
Common TypeScript record and alert model

Map, KPIs, charts, table and detail panel

- Current release: deterministic fictional SIMULATED records plus SCHEDULED station references.
- No live Malaysian provider feed is connected. A failed live source would stay unavailable or stale.
- Rail records are stations, not fabricated train positions.

Interface design

Dashboard hierarchy gives essential information before detail.



- Header search supports the primary locate-a-bus task.
- KPIs and filters update from the same data subset.
- Map selection and table rows reveal the same detail model.
- The table provides an alternative to map interaction.

HCI principles and standards

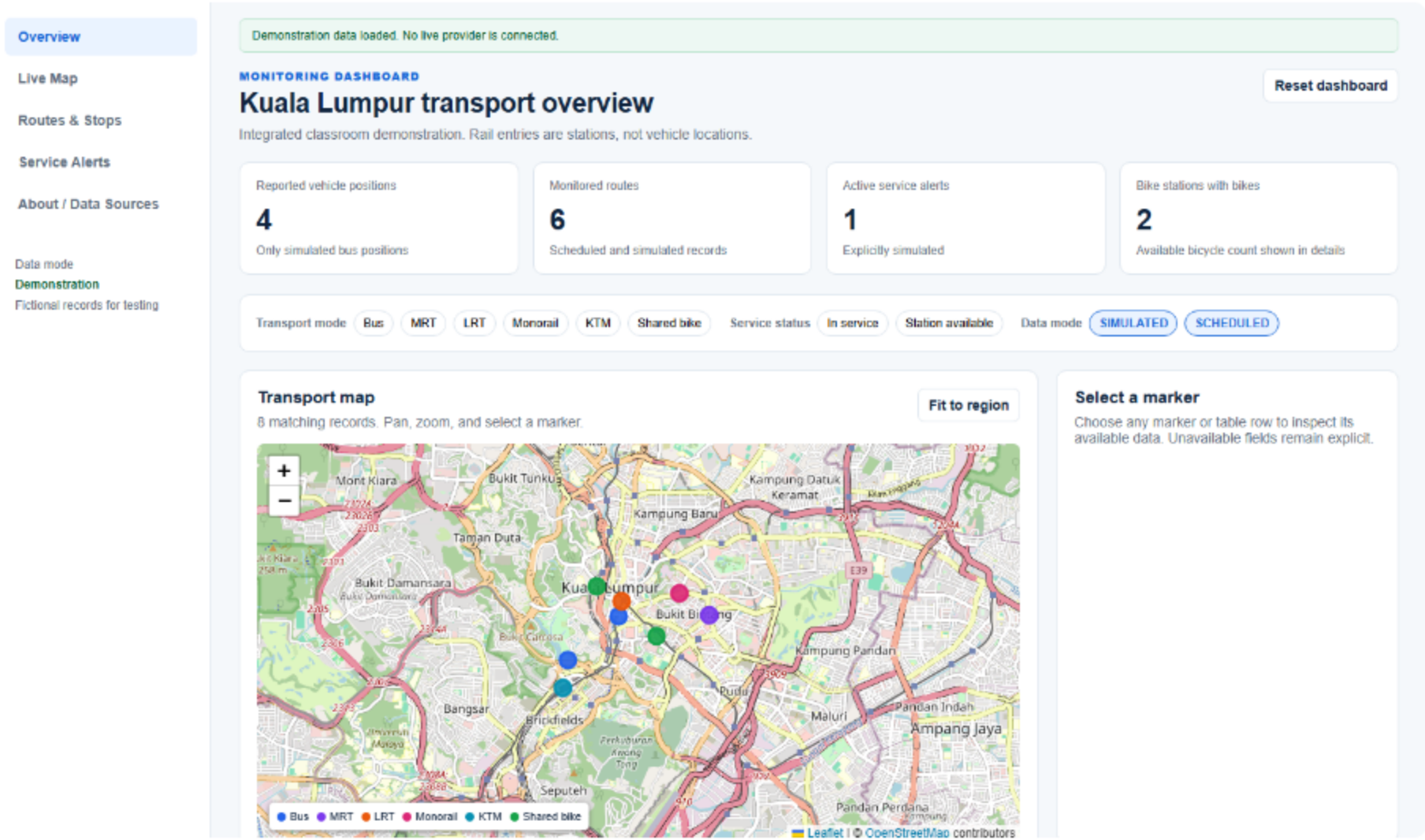
Seven design principles guide the visible interactions.

Principle	Interface evidence
Visibility and feedback	Status message, timestamps, selected filters and alert state.
Consistency and affordance	Shared labels, pressed filter state and recognisable controls.
Control and recovery	Reset, clear search, close alert and empty-state recovery.
Cognitive load	KPI-first hierarchy and selected detail on demand.
Accessibility inspection	Focus rings, semantic controls, labels and non-map table.

Reference frame: ISO 9241-210, ISO 9241-11 and WCAG 2.2. The inspection does not claim full conformance.

Functional website demonstration

Actual screenshot from the local implementation.



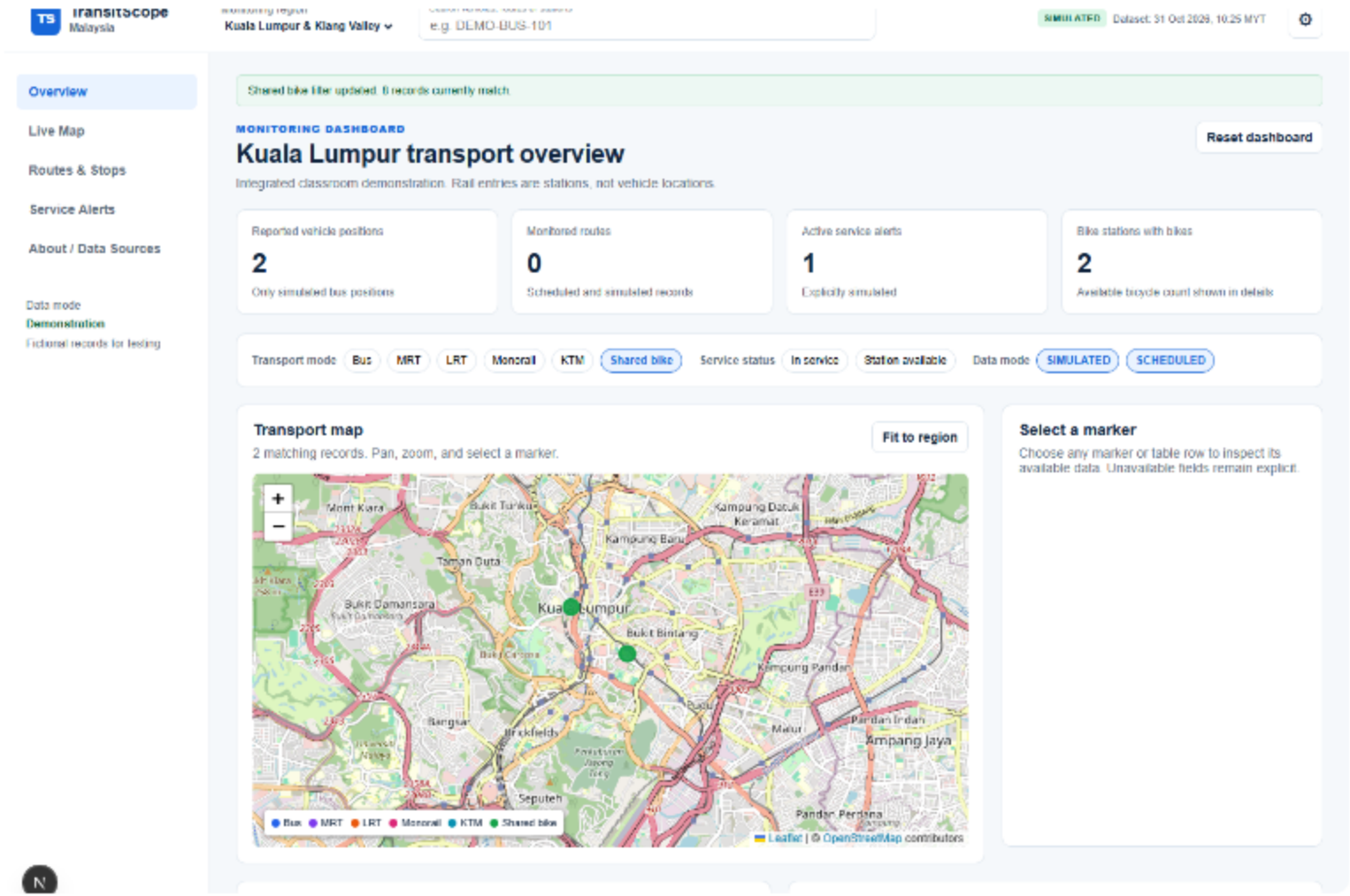
Data mode
SIMULATED

Demo records are fictional.
No live provider is connected.

Map tiles: © OpenStreetMap contributors

Interactions and data visualisation

Filters update the map, KPI values, chart and results list together.



- Search by identifier, route or station
- Transport-mode and status filters
- Marker and table-row selection
- Reset dashboard
- Alert inspection
- Data mode visible in every record

Usability testing methodology

Planned study uses task performance, SUS and an accessibility inspection.

- Aim for at least five representative commuters or students where feasible.
- Record only anonymous ID, device, task result, timing, errors, observations and comments.
- Measure completion rate, successful-task mean time, errors per task attempt and SUS.
- Use the standard ten-question SUS and its published scoring method.

Privacy: no names or contact details in the results sheet. Consent precedes data collection.

Results and HCI metrics

Participant evaluation remains pending.

No participant attempts have been collected.
Task completion rate, completion time, error rate and SUS must remain blank until actual testing occurs.

- Implemented inspection evidence: focus visibility, labels, status feedback, recovery action and responsive breakpoints.
- Remaining: participant results, screen-reader check, formal contrast audit and deployed-device checks.

Discussion and improvements

The current implementation addresses anticipated risks; participant evidence is still required.

Observed / anticipated issue	Implemented change	Evidence still needed
Potential misunderstanding	Simulation disclosure and data mode badge	Validate comprehension in Task A
Filter recovery	Reset dashboard and empty-state restore	Measure errors and backtracking
Map accessibility	Alternative table list and focus indicators	Screen-reader and keyboard testing

One implemented improvement: persistent source transparency prevents demo data from appearing live. Before/after participant evidence remains pending.

Conclusion and reflection

A functional local dashboard establishes a transparent baseline for evaluation.

- Search, mode filtering and alert inspection implement the three predefined tasks.
- The system uses a consistent data model across map, indicators, charts and table.
- The main limitation is the absence of verified live Malaysian transport feeds.
- Next steps: recruit participants, analyse SUS and task metrics, then deploy and record the presentation.

Website, video and references

Submission links stay editable until the student completes deployment and recording.

Website URL: [PENDING PUBLIC DEPLOYMENT]
Recorded Presentation: [INSERT ACTUAL VIDEO LINK]

References (APA 7):
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